

Slide 3 Reading Notes

Analytical approach | Meds Consulting LLC

Opening

This slide explains the analytical approach. The goal is to move from data checks into methods that can support decisions for NHN.

The approach has three parts: regression analysis, customer segmentation, and market basic analysis. Together, these methods connect statistical evidence with operational and financial decisions.

1. Why this approach fits the problem

Readmissions do not come from one single factor. They reflect clinical risk, prior utilization, discharge setting, care-transition activity, and financial exposure. Because these drivers overlap, the analysis needs more than a simple comparison of averages.

Regression helps explain which factors are associated with readmission after accounting for overlap. Segmentation turns those findings into patient groups that teams can act on. Market basic analysis adds context and helps connect clinical results to business scenarios.

2. Model evidence

The chart on the right summarizes model performance. The expanded clinical logistic model improves ROC-AUC from about 0.71 to about 0.74. It also improves recall and F1 score compared with the baseline model.

This does not mean the model is perfect. It means the expanded model gives NHN a stronger basis for risk-tier decisions than a simple baseline model.

Metric	Baseline	Expanded
Accuracy	65.7%	68.3%
Recall	50.1%	55.3%
F1 score	56.0%	60.3%
ROC-AUC	70.7%	74.0%

3. Regression analysis

Regression analysis will quantify which clinical, utilization, discharge, and follow-up variables are associated with readmission. This is useful because it controls for overlap across risk factors.

The output should remain explainable for leadership and clinical teams. We can use odds ratios, feature importance, or predicted risk scores to show which drivers matter most.

4. Customer segmentation

Customer segmentation groups patients into action pools. In this healthcare context, that means grouping patients by chronic burden, prior admissions, discharge setting, follow-up status, and other risk

indicators.

The goal is to identify which patients should receive additional care coordination first. This step makes the analysis practical for operations teams.

5. Market basic analysis

Market basic analysis places the results in context. We compare NHN patterns across service lines and discharge types, then connect trend patterns to financial scenarios.

This helps leadership distinguish broad readmission pressure from specific performance gaps and helps set realistic improvement targets.

Transition cue

After explaining the methods, move to the business insights that come from applying these methods.